

# Triangle Validity and Type

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## DESCRIPTION

Given three positive integers  $a$ ,  $b$ , and  $c$  representing side lengths of a triangle:

1. Check if they can form a valid triangle ( $a + b > c$ ,  $a + c > b$ , and  $b + c > a$ ).
2. If valid, check whether all three sides are equal (Equilateral: `equilateral`), exactly two sides are equal (Isosceles: `isosceles`), or all three sides are different (Scalene: `scalene`).
3. If they cannot form a triangle, output `invalid`.

## INPUT FORMAT

The input contains three positive integers  $a$ ,  $b$ , and  $c$ , separated by spaces.

## OUTPUT FORMAT

Print one of the four words (`equilateral`, `isosceles`, `scalene`, or `invalid`), followed by a newline.

## SAMPLE

| Sample Input | Sample Output |
|--------------|---------------|
| 5 5 8        | isosceles     |